SECP3133 — HIGH PERFORMANCE DATA PROCESSING

Assignment 1: Academic Writing

Semester II 2025/2026 | Faculty of Computing, UTM Johor Bahru

Subject	SECP3133 Pemprosesan Data Berprestasi Tinggi (High Performance Data Processing)
Task	Assignment 1: Academic Writing (Individual)
Weightage	10% of Total Course Assessment
Deadline	Monday, 27 April 2026 — Submit via e-Learning
File Format	PDF | Filename: YourName.pdf (e.g., AhmadFirdaus.pdf)
Submission Portal	e-Learning Platform (upload PDF)

1. Assignment Overview

Welcome, everyone. Let us go through Assignment 1 for SECP3133 — High Performance Data Processing. This is an individual assignment worth 10% of your total course marks. Understanding what is expected and starting early will make a significant difference to the quality of your work and your final grade.

This assignment asks you to write a short academic paper on a topic related to high-performance data processing. The expectation is not merely to summarise what you read — you must demonstrate that you understand the technology, can compare it with older or alternative approaches, and can think critically about its strengths and limitations.

What You Will Develop

By completing this assignment, you will build the following skills, which are highly valued both academically and in industry:

- Academic writing — producing a properly structured, well-cited scholarly paper
- Critical thinking — comparing technologies and forming evidence-based opinions
- Research skills — finding and evaluating high-quality academic sources (Scopus/WoS)
- Information synthesis — combining ideas from multiple papers into a coherent argument
- Digital literacy — responsibly using databases, academic tools, and AI assistants

This assignment is directly linked to the core learning outcomes of HPDP, which centres on understanding modern data processing technologies, architectures, and performance strategies. You will explore real tools and systems used by leading organisations globally — and that is what makes this assignment both relevant and exciting.

2. Topic Selection

You must choose ONE topic from the list provided. There are topics across a broad range of high-performance data processing domains, each paired with a specific tool or technology you should focus on.

How to Choose a Good Topic

- Choose a topic you are genuinely curious about — genuine interest produces better papers
- Verify that enough academic literature exists before committing (check Scopus/WoS first — aim to find at least 10 relevant papers)
- Look for a topic where a clear before-and-after comparison is possible (e.g., Hadoop MapReduce vs. Apache Spark)
- Avoid topics that are too broad — narrow your focus to a specific aspect, use case, or performance dimension

> **TIP:** Search your topic on Scopus/WoS-indexed. If you can find at least 10 relevant Scopus/WoS-indexed papers, you have a solid foundation. If you struggle to find references, reconsider your topic.

Available Topics

Select ONE topic from the table below. You will write your academic paper specifically on the tool or technology indicated in the right column.

No.	Topic Area	Tool / Technology Focus
1	High-Performance Computing	Hadoop / Cloudera
2	Distributed Software	Apache Hadoop
3	Distributed Software	Apache Spark
4	In-Memory Systems	SAP HANA
5	Cloud Computing	AWS (Amazon Web Services)
6	Cloud Computing	Google Cloud Platform
7	In-Database Systems	SAS
8	Cluster Computing	The Virtual SMP Alternative (ScaleMP)
9	Cluster Computing	Hadoop
10	Grid Computing	Hadoop
11	GPU Clusters	NVIDIA CUDA
12	Big Data Analytics	Select any platform
13	Architecture for Data Analytics	Select any platform and compare with previous studies
14	Real-Time Analytics Platforms	Apache Drill
15	Stream Analytics	Apache Spark Streaming
16	NoSQL Databases	MongoDB
17	Storage Area Network	Google File System (GFS)
18	Data Models of NoSQL Databases	Cover all models (e.g., document model in MongoDB)
19	Columnar Databases	Apache HBase
20	Graph Databases	Neo4j
21	NewSQL Systems	Clustrix

No.	Topic Area	Tool / Technology Focus
22	Data-Centric AI Systems	TensorFlow / PyTorch
23	Generative AI in Data Processing	ChatGPT / Gemini (LLMs in data workflows)
24	Modern Data Architectures	Databricks (Lakehouse / Data Mesh)
25	Cloud-Native Architectures	Docker / Kubernetes
26	Serverless Computing	AWS Lambda
27	Edge Computing Systems	Edge AI devices
28	Quantum Computing & Future Trends	IBM Quantum

3. Academic Paper Structure

Your paper must follow a proper academic structure. Think of it like a well-constructed building: each section has a specific purpose, and together they form a complete, coherent argument. The sections below are all mandatory. Missing any component will cost you marks under the Structure criterion.

3.1 Title

Your title is the first thing a reader sees, and it must accurately and specifically reflect your paper's content. Avoid vague or overly general titles.

- Weak title: “Apache Spark” (too vague — what about it?)
- Strong title: “Apache Spark vs. Hadoop MapReduce: A Comparative Analysis of In-Memory Processing Performance for Big Data Workloads”

> **TIP:** A good title typically includes: (1) the technology name, (2) the comparative angle, and (3) the application domain. Aim for 10–15 words.

3.2 Abstract

The abstract is a concise summary of your entire paper (typically 150–250 words). A well-written abstract answers four questions:

1. What is the topic or problem being addressed?
2. Why is it important or relevant?
3. What does this paper discuss or analyse?
4. What are the main findings or conclusions?

> **IMPORTANT:** Write your abstract LAST — after the rest of the paper is complete. A common mistake is writing an abstract that only says what the paper is 'about' without mentioning findings. Make sure yours includes all four elements.

3.3 Keywords (3–5 keywords)

Keywords are specific terms that capture the core topics of your paper. They help readers and search engines find your work. Choose 3 to 5 terms that are precise and widely used in your field.

Example (for a paper on Apache Spark): Keywords: Apache Spark, distributed computing, in-memory processing, big data analytics, performance benchmarking

3.4 Introduction

The introduction gradually guides the reader from a broad context to your specific topic. A strong introduction includes the following elements in order:

5. General background — What is the broader landscape? (e.g., the explosive growth of data in modern enterprises)
6. Problem statement — What challenge or gap exists? (e.g., traditional batch processing is too slow for real-time analytics)
7. Motivation — Why does this topic matter? Who benefits from solving this problem?
8. Paper objective — What does this paper do? (e.g., 'This paper analyses and compares Apache Spark and Hadoop MapReduce...')
9. Paper structure — Briefly describe how the rest of the paper is organised

> **TIP:** Use the Manchester Phrasebank for academic sentence starters: https://www.phrasebank.manchester.ac.uk/ — it is free and covers all paper sections.

3.5 Main Body (Including Comparative Analysis)

This is the most substantial section of your paper and carries the highest marks. It should contain two key components:

Content Discussion

- Explain the technology: What is it? How does it work?
- Describe its architecture with a labelled diagram if possible
- Discuss key features, performance characteristics, and design principles
- Include a real-world case study: a company or system that uses this technology (e.g., Netflix, Uber, Amazon)

Comparative Analysis (MANDATORY — Highest Weighted Criterion)

This is what separates an excellent paper from an average one. You MUST compare your chosen technology with at least one earlier or alternative system. Show what problem the older approach had and how the new technology addresses it.

Examples of meaningful comparisons:

- Apache Spark (modern) vs. Hadoop MapReduce (traditional) — processing speed, memory use, ease of programming
- MongoDB (NoSQL) vs. MySQL (relational SQL) — schema flexibility, horizontal scalability, query expressiveness

- GPU computing (parallel) vs. CPU computing (sequential) — throughput for deep learning workloads

Present your comparison in a structured table for clarity. Example structure:

Criteria	Traditional System (e.g., Hadoop MapReduce)	Modern System (e.g., Apache Spark)
Processing Model	**Disk-based (MapReduce iterations)**	**In-memory processing (RDDs)**
Speed	**Slower due to high disk I/O overhead**	**Up to 100x faster for iterative jobs**
Ease of Use	**Java-heavy; complex API**	**Supports Python, Scala, R, and SQL**
Fault Tolerance	**Data replication on HDFS**	**RDD lineage graph-based recovery**
Best Use Case	**Simple large-scale batch jobs**	**Batch + real-time streaming workloads**
Memory Requirement	**Low (disk-based)**	**Higher RAM required**

Note: The table above is a template example only. Customise your own comparison table based on your chosen topic, and always support each point with a reference.

3.6 Conclusion

The conclusion closes your paper with a synthesis of what was discussed. It should NOT introduce new information or new arguments. A strong conclusion:

- Summarises the key points from the main body
- States the main finding or insight from your comparative analysis
- Identifies limitations of the reviewed technology (if applicable)
- Suggests future research directions or emerging developments

> **IMPORTANT:** Do not copy sentences directly from your introduction or body into the conclusion. Rephrase your key points using fresh language to demonstrate genuine understanding.

3.7 References

List all sources cited in your paper in APA style. Every in-text citation must have a matching reference entry, and every reference must appear somewhere in the text.

APA 7th Edition example for a journal article:

Li, J., Othman, M. S., Ying, X., Hassan, D. S. M., Chen, H., & Mi Yusuf, L. (2026). Adaptive NetFlow IIoT intrusion detection with deep transfer learning, genetic optimization, and ensemble methods for network management. *IEEE Transactions on Network and Service Management, 23*, 681–698. https://doi.org/10.1109/TNSM.2025.3617765

4. Formatting Guidelines

Formatting is part of your assessment. A well-formatted paper shows professionalism and makes your content easier to evaluate. The formatting criteria in the rubric carry 10 marks, so take them seriously.

Requirement	Specification
Page Limit	**Maximum 3 pages (paper content only; log book is additional)**
Paper Template	**IEEE paper template (two-column format is standard for IEEE)**
Font & Size	**Times New Roman or Arial, 10–12pt (follow the IEEE template)**
Line Spacing	**Single-spaced within paragraphs (IEEE standard)**
Margins	**Standard IEEE margins (~1 inch / 2.54 cm on all sides)**
Language	**Formal academic English only — no contractions, informal expressions, or slang**
Paragraph Style	**Justified alignment; write in paragraph form in the main body (not bullet points)**
Figures & Tables	**Label all visuals with numbered captions and cite their source (e.g., 'Figure 1: Spark Architecture (Zaharia et al., 2016)')**
File Format	**Submit as PDF only. Filename must be: YourName.pdf**

> **IMPORTANT:** Papers exceeding the page limit or not following the template will lose marks under the Formatting criterion. Quality and conciseness are more important than length.

5. Reference Requirements

Academic references demonstrate that your work is grounded in credible, peer-reviewed research. For this assignment, the quality of your sources matters as much as the quantity.

Requirements

- Minimum of 7 references
- ALL references must come from Scopus-indexed journals/conferences OR Web of Science (WoS) indexed journals
- Blogs, Wikipedia, YouTube, random websites, and non-indexed sources are NOT acceptable
- Use APA 7th Edition citation style throughout

> **IMPORTANT:** Using non-Scopus/WoS sources will cost you marks under the References criterion. Always verify that your source is indexed before including it.

Where to Find Valid Academic References

10. Scopus — www.scopus.com (access via UTM library portal)
11. Web of Science — www.webofscience.com (access via UTM library portal)
12. Google Scholar — scholar.google.com (use to discover papers, then verify Scopus/WoS indexing)

13. IEEE Xplore — ieeexplore.ieee.org (most IEEE papers are Scopus-indexed)
14. ACM Digital Library — dl.acm.org

> **TIP:** On Google Scholar, papers with high citation counts are usually more credible and relevant. Once you find a paper, check if it is Scopus or WoS indexed before adding it to your references. Scopus lists this on the journal/conference page.

APA Citation Format Examples

In-text citation examples:

- Single author: (Dean, 2008) or Dean (2008) demonstrated that...
- Two authors: (Dean & Ghemawat, 2008)
- Three or more authors: (Zaharia et al., 2016)

Reference list examples:

Journal article: Li, J., Othman, M. S., Ying, X., Hassan, D. S. M., Chen, H., & Mi Yusuf, L. (2026). Adaptive NetFlow IIoT intrusion detection with deep transfer learning, genetic optimization, and ensemble methods for network management. *IEEE Transactions on Network and Service Management, 23*, 681–698. https://doi.org/10.1109/TNSM.2025.3617765

Conference paper: Samsinar, Wahyono, T., Setyawan, I., Karwur, F. F., & Othman, M. S. (2025). Breast cancer detection with CNNs: A systematic review of datasets, challenges, and accuracy. In *Proceedings of the 2025 4th International Conference on Creative Communication and Innovative Technology: Empowering Transformative Mature Leadership: Harnessing Technological Advancement for Global Sustainability (ICCIT 2025)*. https://doi.org/10.1109/ICCIT65724.2025.11166761

6. Plagiarism Policy

Plagiarism means using someone else's ideas, words, or work without properly acknowledging them. This is a serious academic offence and includes copying from internet sources, other students, or even your own previous work without citation.

> **IMPORTANT:** ANY plagiarised work will receive ZERO (0) marks. There are no exceptions, no warnings, and no opportunity to resubmit.

Turnitin Requirement

- A Turnitin similarity report MUST be submitted together with your paper
- Turnitin compares your submission against billions of web pages, academic journals, and previously submitted work
- A similarity score alone does not indicate plagiarism — properly quoted and cited text is acceptable
- High similarity in continuous, uncited chunks is the red flag to avoid

> **TIP:** Aim for a Turnitin similarity score below 20%. If your score is higher, review the flagged sections and paraphrase them in your own words, ensuring all sources are properly cited.

Responsible Use of AI Tools

You may use AI tools (such as ChatGPT, Gemini, or Copilot) to assist with understanding concepts, brainstorming, or improving your writing. However, all AI prompts used must be documented in your Log Book. Using AI to generate your entire paper without genuine understanding and critical engagement is not acceptable and may be detected by your lecturer.

7. Log Book Requirement

A log book is a written record of your work process — think of it as a research journal or project diary. It demonstrates to your lecturer that you engaged with the material genuinely over a period of time, rather than producing everything the night before submission.

What to Record

- Your full name and student ID
- Dates of each research and writing session
- Search keywords used in each session (e.g., 'Apache Spark performance benchmark 2022 Scopus')
- Databases or search engines used (e.g., Scopus, Google Scholar, IEEE Xplore)
- AI prompts used, if applicable (e.g., 'Asked ChatGPT to explain what in-memory processing means in simple terms')
- A brief description of what you accomplished in each session

Example Log Book (you may use a table format like this):

Date	Search Keywords	AI Prompts Used	Task Description
5 April 2026	**Apache Spark performance Scopus**	**None**	**Searched Scopus for papers on Spark performance. Found 12 relevant articles. Selected 7 to read.**
7 April 2026	**Spark vs Hadoop comparison benchmark**	**Explain what in-memory processing means simply**	**Read 5 papers and took notes on key differences. Started drafting the comparison table.**
10 April 2026	**Apache Spark architecture cluster**	**None**	**Drew architecture diagram using draw.io. Wrote the main body section.**
14 April 2026	**APA citation journal example**	**Help me paraphrase this sentence: [paste]**	**Revised the introduction and conclusion. Formatted all references in APA style. Ran Turnitin.**

> **TIP:** Submit your log book as additional pages at the end of your PDF, after the reference list. A complete and detailed log book earns the full 10 log book marks.

8. Strategy to Score High Marks

Many students submit papers that simply describe a technology — they typically receive average marks. Students who score in the excellent range do something different: they analyse, compare, and critically evaluate. Here is exactly what you need to do to maximise your marks.

8.1 Technology Comparison: Before vs. After

Never describe just one technology in isolation. Compare it with the system it replaced or competes with. Demonstrate what was inadequate about the old approach and how the new technology addresses those limitations.

Example of a strong comparative discussion:

"Prior to Apache Spark, Hadoop MapReduce was the dominant framework for large-scale data processing. However, MapReduce required writing intermediate results to disk between processing stages, resulting in significant I/O overhead and latency. Apache Spark addressed this limitation through its Resilient Distributed Dataset (RDD) abstraction, enabling in-memory computation and reducing the number of disk read/write operations dramatically. Benchmarks by Zaharia et al. (2016) indicate that Spark achieves up to 100x faster performance for iterative machine learning workloads compared to MapReduce."

8.2 Include an Architecture Diagram

A well-labelled architecture diagram communicates complex systems clearly and demonstrates your understanding of how the technology is designed. The diagram also counts towards your visuals criterion.

- Create your diagram using draw.io (free), Microsoft Visio, or Lucidchart
- Label all components clearly (e.g., Master Node, Worker Node, Executor, Driver Program)
- Add a proper caption: 'Figure 1: Apache Spark Cluster Architecture (adapted from Li et al., 2026)'
- Always refer to the diagram in your text — do not insert it without discussion

8.3 Include a Real-World Case Study

Grounding your discussion in a real-world example makes your paper more credible and engaging. Choose a well-known company or system that uses your chosen technology and explain how it applies it.

- Netflix — uses Apache Spark for personalised content recommendations at petabyte scale
- Uber — uses Apache Kafka for real-time streaming of millions of ride events per second
- Amazon — uses DynamoDB (NoSQL) to manage billions of product records with millisecond latency
- Tesla — uses edge computing and GPU processing for real-time autonomous vehicle decision-making

> **TIP:** Search 'Company name + technology + case study + Scopus' to find peer-reviewed papers about real-world implementations. This significantly strengthens your paper's credibility.

8.4 Write Critically, Not Just Descriptively

The rubric rewards critical discussion, meaning you must evaluate strengths, weaknesses, limitations, and trade-offs — not just describe what a technology is.

Compare these two approaches:

Descriptive (average marks): *'Apache Spark is a distributed computing framework that supports in-memory processing.'*

Critical (excellent marks): *'While Apache Spark significantly outperforms Hadoop MapReduce for iterative workloads due to in-memory computation, it imposes higher memory requirements and may be cost-prohibitive for simple batch processing on memory-constrained clusters. Zaharia et al. (2016) report up to 100x speed improvement for machine learning tasks, but this advantage diminishes when the dataset size exceeds available RAM, forcing Spark to spill data to disk.'*

8.5 Common Mistakes to Avoid

- Writing informally — use formal academic language at all times; avoid 'I think', 'nowadays', and contractions
- Using non-Scopus/WoS sources — verify every reference before including it
- Missing or shallow comparative analysis — this is the most heavily weighted criterion (10 marks)
- Submitting without a Turnitin report — automatic deduction
- Exceeding the 3-page limit — marks are deducted under Formatting
- Unlabelled figures or tables — all visuals need captions and citations
- Incomplete log book — a vague log book wastes all 10 log book marks
- Late submission — marks are deducted per the rubric timeliness criterion
- Abstract written without findings — must summarise what you found, not just what the paper is about

9. Assessment Rubric

The rubric below shows exactly how your paper will be assessed. Study each criterion carefully and use it as a reference checklist when reviewing your work before submission. Total marks: 100 (scaled to 10% of course assessment).

> **IMPORTANT:** Note: The rubric references the IEEE paper template and page lengths of 4–5 pages. Please confirm with your lecturer which template and page length apply to the current semester, as the assignment brief specifies a 3-page limit with APA referencing.

Criteria	Excellent (7–10)	Fair (4–6)	Poor (0–3)	Weightage
Structure (Full Mark: 5)	**ALL required components present: title, abstract, keywords, introduction, content with comparative analysis, conclusion, references**	**SOME required components included**	**Most components are missing**	**5 marks**
Organisation & Writing Style — Organisation (Full Mark: 10)	**Well-organised; clear and logical presentation throughout**	**Moderate organisation and presentation**	**Poorly organised; points are disjointed and jumpy**	**5 marks**
Organisation & Writing Style — Readability (Full Mark: 10)	**Easy to read; proper paragraph writing; clear and concise**	**Moderate writing style**	**Difficult to read; confusing writing**	**5 marks**

Criteria	Excellent (7–10)	Fair (4–6)	Poor (0–3)	Weightage
Content: Relevance to Topic (Full Mark: 35)	Paper adequately ties to the selected topic	Paper moderately ties to topic	Paper does not tie to topic	2 marks
Content: Title Quality	Title adequately represents the content	Title moderately represents content	Title does not represent content	2 marks
Content: Abstract Quality	Abstract accurately reflects the full content of the paper	Abstract moderately reflects content	Abstract does not reflect content	2 marks
Content: Clear Purpose	Purpose of the article is stated clearly	Purpose is stated moderately	Purpose is not clear	2 marks
Content: Introduction	Introduction adequately provides necessary background information	Introduction moderately provides background	Introduction poorly provides background	2 marks
Content: Comparative Analysis (HIGHEST WEIGHT)	Comparative analysis of previous studies is thorough and includes ALL required comparative points	Comparative analysis includes SOME required points	Comparative analysis is poor; very few required points included	10 marks
Content: Depth of Discussion	Content of selected previous studies is discussed thoroughly and critically	Content is discussed moderately	Content discussed very generally (shallow)	8 marks
Content: Graphics & Visuals	All figures, tables, and photos are appropriate, adequate, and properly cited	Some graphics are appropriate and adequate	Very few graphics are appropriate	4 marks
Content: Conclusion & References	Conclusions are accurate and supported by content; references are relevant	Conclusions and references are acceptable	Conclusions are weak; references not relevant	3 marks
Attractiveness (Full Mark: 5)	Paper is engaging and interesting to the reader	Paper is acceptable to the reader	Paper is dull or boring	5 marks
Log Book (Full Mark: 10)	Complete and detailed record of the research and writing process	Record is acceptable	Record is incomplete	10 marks
Formatting: Template (Full Mark: 10)	Fully formatted according to IEEE paper template	Partly formatted according to IEEE template	Messy; not following IEEE template	5 marks
Formatting: Citations	References properly cited throughout text following IEEE/APA guidelines	Citations have frequent inconsistencies or errors	No in-text citations	3 marks
Formatting: Length of Paper	Length = 4 pages (Rubric standard; check with lecturer for current requirement)	Length = 5 pages	Length > 5 pages	2 marks

Criteria	Excellent (7–10)	Fair (4–6)	Poor (0–3)	Weightage
Mechanics (Full Mark: 10)	**Paper is free from grammatical or spelling errors**	**Some grammar or spelling errors but acceptable**	**Very poor; too many grammar and spelling errors**	**10 marks**
Timeliness (Full Mark: 5)	**Submitted on time**	**Submitted one day late**	**Submitted more than one day late**	**5 marks**
Presentation (Full Mark: 10)	**Holds attention of entire audience; direct eye contact; speaks with volume variation and inflection to emphasise key points**	**Consistent eye contact; satisfactory variation in volume and inflection**	**No eye contact; reads entirely from notes; monotonous or low-volume delivery**	**10 marks**

TOTAL: 100 marks (scaled to 10% of total course assessment)

10. Step-by-Step Guide to Completing This Assignment

Follow this structured three-week plan. Do not attempt to complete everything in one sitting — spreading your work over time leads to a much better paper and a much lower stress level.

Week 1 — Planning & Research

15. Select your topic. Make sure it genuinely interests you and that you can find enough Scopus/WoS references.
16. Search for at least 10–15 academic papers on your topic using Scopus and Google Scholar.
17. Identify papers that compare your technology with an older or alternative system.
18. Read abstracts and conclusions first. Then read the full papers that are most relevant.
19. Begin your log book immediately. Record all search keywords, databases used, and dates.

Week 2 — Writing the Paper

20. Create a detailed outline listing all the main points for each section.
21. Write the Introduction, then the Main Body, then the Conclusion.
22. Build your comparison table using structured criteria and cited data.
23. Create or adapt an architecture diagram. Label all components and cite it.
24. Add your references to the reference list in APA format as you write.
25. Continue updating your log book with each writing session.

Week 3 — Polishing & Submission

26. Write the Abstract (do this last — summarise what you actually wrote).
27. Proofread thoroughly for grammar, spelling, punctuation, and consistency.
28. Verify all in-text citations match the reference list.
29. Apply the IEEE paper template format.

30. Run Turnitin. Review flagged sections and paraphrase as needed. Aim below 20% similarity.
31. Complete your log book. Ensure all sessions are recorded.
32. Compile everything into a single PDF: paper + log book + Turnitin report.
33. Rename the file: YourName.pdf and submit via e-learning before Monday, 27 April 2026.

11. Final Submission Checklist

Before you click the submit button, go through every item on this checklist. Do not submit until you can tick every box with confidence.

✓	Checklist Item
[ ]	**Paper has a clear, specific, and informative Title (10–15 words)**
[ ]	**Abstract is written (150–250 words) and covers: topic, importance, discussion, and main findings**
[ ]	**3 to 5 Keywords are listed below the abstract**
[ ]	**Introduction provides background, problem statement, motivation, and paper objective**
[ ]	**Main Body contains thorough discussion of the chosen technology**
[ ]	**Comparative analysis is included — comparing the old system vs. the modern one**
[ ]	**A real-world case study or application example is discussed with a reference**
[ ]	**An architecture diagram is included, labelled, and cited in the caption**
[ ]	**Conclusion summarises findings and suggests future research directions**
[ ]	**Minimum 7 references listed, ALL from Scopus or WoS indexed sources**
[ ]	**All references are formatted correctly in APA 7th Edition style**
[ ]	**All in-text citations are correctly formatted and match the reference list**
[ ]	**Paper is formatted according to the IEEE paper template**
[ ]	**Paper does NOT exceed the page limit (confirm with lecturer: 3 pages per brief)**
[ ]	**All figures and tables have numbered captions and source citations**
[ ]	**Grammar, spelling, and punctuation have been thoroughly proofread**
[ ]	**Turnitin similarity report is prepared (aim: below 20%) and ready to attach**
[ ]	**Log book is complete: dates, keywords, AI prompts, task descriptions, and student ID**
[ ]	**PDF file is named correctly: YourName.pdf**
[ ]	**Submission uploaded via e-learning BEFORE Monday, 27 April 2026**

A Final Word of Encouragement

This assignment is an opportunity to explore technologies that are genuinely shaping the future of computing and data science. Apache Spark, Kafka, Kubernetes, Neo4j, IBM Quantum — these are not just academic concepts. They are the systems powering the world's most innovative companies right now.

Approach this paper with genuine curiosity. Read the research, form your own informed opinion, and write with confidence. A paper that shows critical thinking and authentic engagement will always stand out far more than one that simply describes.

Start early. Plan well. Write with purpose. Good luck — we look forward to reading your work.

Academic Writing Resource: https://www.phrasebank.manchester.ac.uk/